



GLOBAL ECONOMICS
GRADE: 10TH
TERM 3 — CATCH-UP ACTIVITY
STANDARD NORMAL, CONTEXTUAL NORMAL, AND
INVERSE NORMAL PROBLEMS (C5–C7)
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TERM 3
2025–2026

Learning objective: Strengthen the use of standard normal, contextual normal, and inverse normal methods to solve probability problems in C5, C6, and C7.

Assessment criteria:

C5. Uses the standard normal distribution and its tables to calculate probabilities.

C6. Uses normal distributions in contextual situations.

C7. Uses inverse normal reasoning to find values and intervals.

Criteria assessment

Each criterion has a maximum score of 14 points. To pass a criterion, students must obtain at least 12 total points. Points earned during the midterm count toward this total. The work must be completed individually in class. The final submission deadline is Friday, June 5.

1. Standard Normal Table Problems (C5)

Let $Z \sim N(0, 1)$. Let $\Phi(z) = P(Z < z)$.

This section trains students to compute standard-normal probabilities using the table.

Questions/Tasks.

a) (6) Direct and complement-rule probabilities.

- 1) Find $P(Z < 1.24)$.
- 2) Find $P(Z > -0.67)$.
- 3) Find $P(Z > 1.18)$.
- 4) Find $P(Z < -0.52)$.
- 5) Find $P(Z < 0.36)$.
- 6) Find $P(Z > 2.05)$.
- 7) Find $P(Z < -1.41)$.
- 8) Find $P(Z > -1.09)$.
- 9) Find $P(Z > 0.73)$.
- 10) Find $P(Z < 1.96)$.
- 11) Find $P(Z < -0.88)$.
- 12) Find $P(Z > -2.17)$.

b) (4) Interval probabilities.

- 1) Find $P(0.42 < Z < 1.37)$.
- 2) Find $P(-0.95 < Z < 1.12)$.
- 3) Find $P(-1.75 < Z < -0.40)$.
- 4) Find $P(-2.10 < Z < 0.65)$.

c) (4) Tail-union probabilities.

- 1) Find $P(Z < -1.25 \text{ or } Z > 1.80)$.
- 2) Find $P(Z < 0.45 \text{ or } Z > 1.70)$.
- 3) Find $P(Z < -2.00 \text{ or } Z > -0.60)$.
- 4) Find $P(Z < -0.30 \text{ or } Z > 2.25)$.

2. Contextual Normal Distribution Problems (C6)

Use

$$Z = \frac{X - \mu}{\sigma}$$

to convert each contextual value into a standard normal value.

Questions/Tasks.

- 1) **(1) Exam scores.** If exam scores follow $X \sim N(75, 10^2)$, find $P(X < 87.4)$ and $P(X > 68.3)$.
- 2) **(1) Delivery times.** If delivery times follow $X \sim N(40, 6^2)$, find $P(X > 47.08)$ and $P(X < 36.88)$.
- 3) **(1) Product weights.** If product weights follow $X \sim N(500, 20^2)$, find $P(X < 507.2)$ and $P(X > 541)$.
- 4) **(1) Commute times.** If commute times follow $X \sim N(35, 8^2)$, find $P(X < 23.72)$ and $P(X > 26.28)$.
- 5) **(1) Customer spending.** If customer spending follows $X \sim N(120, 15^2)$, find $P(X > 130.95)$ and $P(X < 149.4)$.
- 6) **(1) Machine lifetimes.** If machine lifetimes follow $X \sim N(1000, 100^2)$, find $P(X < 912)$ and $P(X > 783)$.
- 7) **(2) Weekly app usage.** If weekly app usage follows $X \sim N(20, 4^2)$, find $P(21.68 < X < 25.48)$ and $P(16.2 < X < 24.48)$.

- 8) **(2) Crop yields.** If crop yields follow $X \sim N(60, 5^2)$, find $P(51.25 < X < 58)$ and $P(49.5 < X < 63.25)$.
- 9) **(2) Battery duration.** If battery duration follows $X \sim N(12, 2^2)$, find $P(X < 9.5 \text{ or } X > 15.6)$ and $P(X < 12.9 \text{ or } X > 15.4)$.
- 10) **(2) Scholarship scores.** If scholarship scores follow $X \sim N(300, 30^2)$, find $P(X < 240 \text{ or } X > 282)$ and $P(X < 291 \text{ or } X > 367.5)$.

3. Inverse Normal Problems (C7)

Use

$$Z = \frac{X - \mu}{\sigma} \quad \text{and} \quad x = \mu + \sigma z$$

to convert between standard normal values and contextual values.

Questions/Tasks.

- 1) **(1) Commute times.** Commute times follow $X \sim N(38, 7^2)$.
- Find x such that $P(X < x) = 0.80$.
 - Find x such that $P(X > x) = 0.10$.
- 2) **(1) Exam scores.** Exam scores follow $X \sim N(72, 9^2)$.
- Find x such that $P(X > x) = 0.80$.
 - Find x such that $P(X < x) = 0.25$.
- 3) **(1) Delivery times.** Delivery times follow $X \sim N(56, 5^2)$.
- Find x such that $P(X < x) = 0.60$.
 - Find x such that $P(X > x) = 0.30$.
- 4) **(1) Product weights.** Product weights follow $X \sim N(500, 20^2)$.
- Find x such that $P(X < x) = 0.10$.
 - Find x such that $P(X > x) = 0.75$.
- 5) **(1) Customer spending.** Customer spending fol-

lows $X \sim N(120, 15^2)$.

- Find x such that $P(X > x) = 0.05$.
 - Find x such that $P(X < x) = 0.70$.
- 6) **(1) App usage.** App usage follows $X \sim N(20, 4^2)$.
- Find x such that $P(X < x) = 0.40$.
 - Find x such that $P(X > x) = 0.90$.
- 7) **(2) Battery life.** Battery life follows $X \sim N(20, 2.5^2)$.
- Find $[a, b]$ such that $P(a < X < b) = 0.90$, with a central interval.
 - Find $[a, b]$ such that $P(X < a) = 0.60$ and $P(X < b) = 0.90$.
- 8) **(2) Machine lifetimes.** Machine lifetimes follow $X \sim N(1000, 100^2)$.
- Find $[a, b]$ such that $P(X < a) = 0.10$ and $P(X < b) = 0.40$.
 - Find $[a, b]$ such that $P(X < a) = 0.25$ and $P(X < b) = 0.75$.
- 9) **(2) Metal rod diameters.** Metal rod diameters follow $X \sim N(10.00, 0.08^2)$.
- Find a and b such that $P(X < a \text{ or } X > b) = 0.10$, with symmetric tails.
 - Find a and b such that $P(X < a \text{ or } X > b) = 0.70$, with $P(X < a) = 0.60$ and $P(X > b) = 0.10$.
- 10) **(2) Crop yields.** Crop yields follow $X \sim N(60, 5^2)$.
- Find a and b such that $P(X < a \text{ or } X > b) = 0.70$, with $P(X < a) = 0.10$ and $P(X > b) = 0.60$.
 - Find a and b such that $P(X < a \text{ or } X > b) = 0.45$, with $P(X < a) = 0.20$ and $P(X > b) = 0.25$.